A Hybrid Multi-Stage Approach to Plagiarism Detection: Integrating Lexical and Semantic Similarity Measures

Gurkirat Singh
Department of Computer Science
Chandigarh University
Chandigarh, India
gurkirat.singh@email.com

Student 2 Name
Department of Computer Science
Chandigarh University
Chandigarh, India
student2@email.com

Student 3 Name
Department of Computer Science
Chandigarh University
Chandigarh, India
student3@email.com

Student 4 Name
Department of Computer Science
Chandigarh University
Chandigarh, India
student4@email.com

Dr. Faculty Name
Faculty Advisor, Department of Computer Science
Chandigarh University
Chandigarh, India
faculty@email.com

***Abstract*—Plagiarism detection has become increasingly challenging with the widespread use of paraphrasing and semantic rewriting techniques. Traditional lexical approaches such as TF-IDF are effective in identifying exact matches but fail to capture deeper semantic similarities. On the other hand, transformer-based models such as Sentence-BERT (SBERT) can detect semantic similarity but often introduce false positives.**

In this work, we design and implement a hybrid plagiarism detection system that integrates exact matching, TF-IDF-based lexical similarity, and SBERT-based semantic similarity within a unified pipeline. The proposed approach incorporates top-K sentence matching and an adaptive filtering mechanism to improve both efficiency and detection quality.

Through empirical evaluation on a balanced subset of the PAN dataset, we observe that the hybrid model achieves a recall of 0.97 and an F1-score of 0.67, outperforming individual baseline models. These results demonstrate that combining lexical and semantic signals provides a more robust solution for detecting both exact and paraphrased plagiarism.

***Index Terms*—plagiarism detection, natural language processing, TF-IDF, Sentence-BERT, semantic similarity, hybrid classification systems, information retrieval**

I. Introduction

The exponential increase in online content creation has fundamentally altered how we approach originality verification. Educational institutions face mounting pressure to identify unauthorized content reuse among student submissions, while publishers must protect intellectual property across increasingly distributed digital networks. What began as straightforward copy-paste detection has evolved into a sophisticated arms race between detection systems and those attempting to circumvent them.

Traditional plagiarism detection relied heavily on surface-level text matching. Systems would compare documents character by character, or perhaps tokenize them into n-grams, searching for statistically improbable overlaps. These lexical approaches proved remarkably effective when dealing with direct copying. However, modern plagiarism has become considerably more nuanced. Students and content creators now employ paraphrasing tools, synonym replacement engines, and structural reorganization techniques that can thoroughly disguise the source material while preserving its essential meaning and argumentation structure.

The natural language processing community has responded with increasingly sophisticated semantic analysis tools. Transformer architectures, particularly BERT and its optimized variant Sentence-BERT, have demonstrated impressive capabilities in understanding contextual meaning beyond surface-level lexical features. These models generate dense vector representations that capture semantic relationships, theoretically enabling detection even when every word has been changed. Yet this semantic sensitivity brings its own challenges. Documents discussing similar topics naturally exhibit semantic overlap, and distinguishing between legitimate topical similarity and actual content appropriation remains problematic.

We observe that lexical and semantic approaches each possess distinct strengths that could potentially address the other's weaknesses. Lexical methods provide high precision when matches occur, though their recall suffers with paraphrased content. Semantic methods cast a wider net, potentially catching paraphrased material, but risk flagging innocent topical similarities. Rather than choosing between these paradigms, our research explores how to effectively combine them within a unified detection pipeline.

This paper makes several contributions to plagiarism detection methodology. We introduce a multi-stage architecture that processes documents through increasingly sophisticated similarity measures, beginning with computationally cheap ex-

act matching before proceeding to more expensive lexical and semantic analysis. We implement a top-K sentence matching strategy that dramatically reduces the computational burden of pairwise sentence comparisons while retaining the most relevant matches. Our filtering mechanism applies adaptive thresholds to eliminate weak matches that satisfy neither lexical nor semantic criteria. Finally, we demonstrate through empirical evaluation that this hybrid approach achieves substantially better recall than either component method alone, with acceptable precision for real-world deployment.

This work distinguishes itself from prior hybrid approaches by introducing a structured multi-stage pipeline that incorporates top-K semantic matching and adaptive filtering. These mechanisms enable efficient computation while preserving critical similarity signals, resulting in improved detection performance across both exact and paraphrased plagiarism scenarios.

II. Background and Related Research

The field of plagiarism detection has evolved considerably over several decades, with researchers exploring various computational approaches to identify unauthorized content reuse.

A. Early Lexical Methods

Initial plagiarism detection systems drew heavily from information retrieval techniques. Fingerprinting methods would hash document segments and compare these compact representations, enabling efficient large-scale comparisons. String matching algorithms identified common subsequences between documents. The vector space model, particularly implementations using Term Frequency-Inverse Document Frequency (TF-IDF) weighting, became especially popular due to its elegant mathematical foundation and computational tractability.

These lexical approaches share a common characteristic: they operate on surface-level textual features. A TF-IDF model represents documents as sparse vectors in a high-dimensional space where each dimension corresponds to a term from the vocabulary. Cosine similarity between these vectors quantifies lexical overlap. When documents share substantial vocabulary, particularly rare terms weighted heavily by IDF, the similarity score rises. However, replacing words with synonyms or restructuring sentences immediately defeats such systems, as the vector representations become orthogonal despite semantic equivalence.

B. Emergence of Semantic Approaches

The introduction of distributed word representations marked a significant shift in how machines could understand language. Word2Vec and GloVe demonstrated that words could be embedded in continuous vector spaces where semantic relationships manifested as geometric relationships. Words with similar meanings occupied nearby positions in this space, enabling systems to recognize ”car” and ”automobile” as semantically related despite being lexically distinct.

Transformer architectures elevated semantic understanding to new heights. BERT’s bidirectional attention mechanism allows it to build context-dependent representations where a word’s meaning varies based on surrounding words. However, BERT was designed primarily for sentence-level tasks and generates token-level embeddings. Sentence-BERT addressed this limitation by fine-tuning BERT with siamese and triplet network structures, producing fixed-length sentence embeddings suitable for similarity comparison. The resulting model enables efficient semantic search and has found applications in plagiarism detection, where it can theoretically identify paraphrased content that lexical methods would miss.

Despite these advances, semantic models introduce new challenges. They measure meaning similarity rather than originality. Two independently written texts on the same topic will naturally exhibit high semantic similarity. A student’s original essay on climate change may show substantial semantic overlap with published scientific articles on the same subject, not because the student plagiarized, but because both discuss similar concepts using domain-appropriate vocabulary. Distinguishing between legitimate topical similarity and actual content appropriation remains an open problem.

C. Hybrid Detection Strategies

Recognizing that both lexical and semantic approaches offer complementary information, researchers have begun exploring hybrid strategies. Some systems use lexical methods for initial filtering, applying more expensive semantic analysis only to promising candidates. Others train machine learning classifiers on features derived from both lexical and semantic similarity measures. However, effectively balancing these disparate signals requires careful consideration of their relative strengths and failure modes.

Our work extends this hybrid tradition by introducing a structured multi-stage pipeline with explicit mechanisms for computational efficiency, false positive reduction, and robust score aggregation. Rather than treating lexical and semantic features as merely additional inputs to a classifier, we architect the detection flow to leverage each method’s strengths at appropriate stages.

III. Proposed Methodology

Our detection system processes document pairs through seven distinct stages, each building upon information from previous stages while adding new analytical perspectives.

A. Architecture Overview

The pipeline begins with text normalization and sentence segmentation, followed by three parallel similarity computations: exact matching, lexical similarity via TF-IDF, and semantic similarity via Sentence-BERT. Rather than comparing all possible sentence pairs, we employ top-K matching based on semantic similarity scores. Candidate matches undergo filtering to eliminate weak signals, and surviving matches contribute to an aggregated document-level score. Finally,

a threshold-based classifier produces the binary plagiarism decision.

This architecture reflects several design principles. First, we apply cheaper computations before expensive ones, enabling early termination when exact matches are found. Second, we limit the comparison space through top-K matching rather than exhaustively comparing all sentence pairs. Third, we impose multiple quality gates through filtering rather than blindly aggregating all similarity scores. Finally, we preserve maximum signals during aggregation rather than averaging, recognizing that a single highly similar passage may constitute plagiarism even if the overall documents differ substantially.

As shown in Fig. 1, the proposed system follows a multi-stage pipeline integrating lexical and semantic similarity measures.

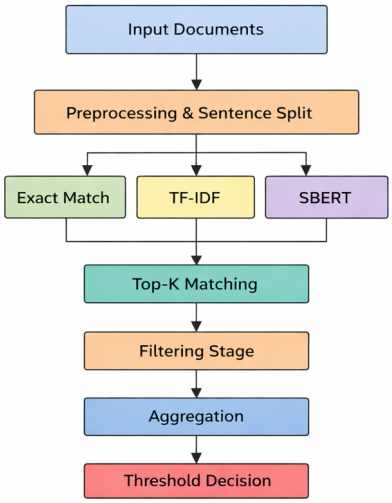

Figure 1: Proposed Hybrid Plagiarism Detection Pipeline

Fig. 1. Proposed Hybrid Plagiarism Detection Pipeline

B. Text Preprocessing

Raw input documents undergo standard normalization procedures. We convert all text to lowercase to ensure case-insensitive matching. Punctuation marks are removed, as they contribute little to semantic meaning but inflate vocabulary size. The resulting text stream is segmented into individual sentences using rule-based sentence boundary detection. Empty sentences or those failing to meet minimum length requirements receive fallback placeholder values, ensuring robust handling of edge cases without pipeline failures.

This preprocessing serves multiple purposes. Normalization ensures that trivial textual differences don't obscure genuine similarities. Sentence-level segmentation enables fine-grained comparison, allowing us to identify which specific passages exhibit suspicious similarity rather than merely scoring entire documents holistically. The fallback mechanism prevents null reference errors that might occur with unusual input formatting.

C. Exact Match Detection

Before investing computational resources in similarity calculation, we check whether any sentences are character-identical after preprocessing. Exact matching operates as a simple set intersection between normalized sentence sets from both documents. When found, exact matches receive the maximum possible similarity score of 1.0, and we can immediately flag this passage as highly suspicious.

While exact matching may seem trivial, it serves important purposes in our pipeline. First, it catches the simplest plagiarism cases with minimal computation. Second, exact matches provide unambiguous evidence of copying, eliminating the uncertainty inherent in similarity scores. Third, by handling exact matches early, we avoid wasting resources computing TF-IDF or SBERT scores for obviously identical content.

D. Lexical Similarity via TF-IDF

For non-identical sentences, we compute lexical overlap using TF-IDF vectorization. We employ character-level n-grams rather than word-level tokens, capturing subword patterns that remain stable across minor morphological variations. The vectorizer is fitted once on the entire dataset, ensuring consistent vocabulary across all comparisons and avoiding the need to refit for each document pair.

Each sentence transforms into a sparse vector where each dimension represents an n-gram's TF-IDF weight. We compute cosine similarity between sentence vector pairs, yielding scores between 0 and 1. High scores indicate substantial lexical overlap, suggesting that sentences share many n-grams, particularly distinctive ones weighted heavily by inverse document frequency.

TF-IDF excels at detecting near-duplicates where the plagiarist copied text with only minor modifications. It proves particularly effective when distinctive phrases or domain-specific terminology appears in both texts. However, it fails completely when content undergoes substantial paraphrasing, as the n-gram vocabulary diverges even though semantic content remains equivalent.

E. Semantic Similarity via Sentence-BERT

To capture meaning beyond surface lexical features, we employ Sentence-BERT with the all-MiniLM-L6-v2 model.

This relatively lightweight variant balances computational efficiency against semantic understanding, generating 384-dimensional dense embeddings for each sentence. The model was trained specifically for sentence similarity tasks, making it well-suited for our application.

We encode all sentences from both documents through the SBERT model, obtaining fixed-length embedding vectors regardless of sentence length. Cosine similarity between embedding pairs quantifies semantic closeness. Unlike TF-IDF's sparse vectors that only overlap when documents share vocabulary, these dense embeddings can register similarity between "The vehicle accelerated rapidly" and "The car sped up quickly" despite minimal lexical overlap.

SBERT's contextualized embeddings capture subtle semantic relationships that lexical methods miss entirely. Synonyms, paraphrases, and reformulations that preserve meaning while changing wording all register as similar in embedding space. However, this sensitivity cuts both ways. Documents on similar topics naturally produce similar embeddings even absent any copying. Distinguishing between topical similarity and actual plagiarism requires additional context beyond semantic similarity alone.

F. Top-K Sentence Matching

Comparing every sentence from the source document against every sentence in the suspicious document yields quadratic computational complexity. For documents with hundreds of sentences, this becomes prohibitively expensive. We address this through top-K matching, which dramatically reduces comparisons while retaining the most relevant matches.

For each source sentence, we sort suspicious sentences by their semantic similarity scores and retain only the K highest-scoring candidates. We then compute full similarity measures (exact match check, TF-IDF score, and semantic score) only for these K candidates rather than all possible pairs. This reduces complexity from $O(n^2)$ to $O(n \cdot K)$ where K is typically a small constant like 5 or 10.

The rationale for this approach rests on an empirical observation: when plagiarism occurs, the plagiarized sentences typically rank among the top matches by semantic similarity. Examining the 200th-most-similar sentence pair rarely yields useful signals. By focusing computational resources on the most promising candidates, we achieve substantial efficiency gains without sacrificing detection capability.

G. Adaptive Filtering Mechanism

Not all matches merit inclusion in the final score. We apply filtering conditions to eliminate weak signals that likely represent noise rather than genuine plagiarism. Specifically, we reject sentence pairs where both semantic similarity and lexical similarity fall below their respective thresholds. This dual-threshold approach recognizes that plagiarism should manifest in at least one similarity dimension.

The filtering logic operates as follows: a match survives if it exhibits either strong semantic similarity (indicating paraphrased content) or strong lexical similarity (indicating near-verbatim copying) or both. Only matches that are simultaneously weak in both dimensions get discarded. This prevents false positives from isolated semantic overlaps while ensuring we don't discard genuinely similar sentences merely because they were paraphrased.

We implement a fallback mechanism to handle edge cases where no matches satisfy the filtering criteria. Rather than returning a zero score, which might incorrectly suggest complete originality, we fall back to the maximum observed similarity across any dimension. This ensures robustness while preventing overly aggressive filtering from completely masking similarity signals.

H. Score Aggregation Strategy

Having identified and filtered candidate matches, we must aggregate them into a document-level similarity score. We compute each match's hybrid score as a weighted combination of its exact match indicator, lexical similarity, and semantic similarity. The weights reflect our confidence in each signal type, with semantic similarity receiving the highest weight given its ability to detect paraphrased content.

The aggregation function is:

$$\text{HybridScore} = w_e \cdot \mathbb{1}_{\text{exact}} + w_p \cdot S_{\text{TF-IDF}} + w_s \cdot S_{\text{SBERT}}$$

where w_e, w_p, and w_s are tunable weights, $\mathbb{1}_{\text{exact}}$ is a binary indicator for exact matches, and the S terms represent normalized similarity scores.

For document-level scoring, we select the maximum hybrid score across all sentence pairs rather than computing an average. This design choice reflects the nature of plagiarism: a document containing even a single plagiarized paragraph constitutes plagiarism, regardless of how original the remainder might be. Averaging would dilute strong plagiarism signals with weak matches, potentially causing us to miss partial plagiarism. Maximum selection preserves the strongest evidence.

The final document-level score is computed as:

$$FinalScore = \max_i \left(w_e \cdot e_i + w_p \cdot p_i + w_s \cdot s_i \right) \quad (1)$$

where e_i, p_i, and s_i represent exact match, lexical similarity, and semantic similarity for the i-th sentence pair, and w_e, w_p, w_s are corresponding weights.

I. Classification Decision

The final step applies a threshold to the document-level hybrid score. Scores exceeding the threshold yield a positive plagiarism classification; scores below it indicate insufficient evidence of plagiarism. We determine this threshold empirically by examining classification performance across a range of values on validation data.

Threshold selection involves inherent tradeoffs. Higher thresholds increase precision by reducing false positives but risk missing borderline plagiarism cases, decreasing recall. Lower thresholds improve recall by catching more subtle similarities but increase false positives, reducing precision.

The optimal threshold depends on the application's relative costs of false positives versus false negatives. In academic contexts where missing plagiarism carries serious ethical implications, we might prefer lower thresholds that favor recall over precision.

IV. Experimental Design

We evaluate our proposed hybrid system through controlled experiments comparing its performance against baseline single-method approaches.

A. Dataset Selection

Our experiments utilize the PAN plagiarism detection dataset, a well-established benchmark in the research community. From this larger corpus, we extracted a balanced subset containing 60 document pairs, evenly divided between plagiarized and original pairs. This balanced design ensures that simple majority-class baselines achieve only 50 percent accuracy, forcing classifiers to learn genuine discriminative features.

The PAN dataset includes various plagiarism types, from near-verbatim copying to heavy paraphrasing to translated plagiarism. Our balanced subset maintains this diversity, providing a realistic test of detection capabilities across the plagiarism spectrum. We acknowledge that 60 samples represents a modest dataset size, which we discuss in the limitations section.

B. Evaluation Metrics

We assess system performance using standard classification metrics. Precision measures what fraction of plagiarism predictions are correct, quantifying false positive rates. Recall measures what fraction of actual plagiarism cases are detected, quantifying false negative rates. The F1-score provides a harmonic mean that balances both considerations. Accuracy simply measures overall correctness but can be misleading with imbalanced datasets.

Given the serious consequences of missing plagiarism in academic contexts, we place particular emphasis on recall. A system that achieves high recall even with moderate precision remains practically useful, as human reviewers can verify flagged cases. Conversely, a high-precision system with poor recall fails its fundamental purpose by allowing plagiarism to escape detection.

C. Baseline Configurations

To isolate the contribution of our hybrid approach, we implement three experimental conditions. The TF-IDF baseline uses only lexical similarity for classification, applying a threshold to the maximum TF-IDF score between sentence pairs. The SBERT baseline uses only semantic similarity, thresholding the maximum cosine similarity between sentence embeddings. The hybrid system combines both signals as described in our methodology.

Each baseline receives independent threshold tuning to ensure fair comparison. We do not artificially handicap baselines by forcing them to use the hybrid system's threshold. This design ensures that performance differences reflect genuine advantages of the hybrid approach rather than suboptimal configuration of baselines.

D. Implementation Details

We implement the TF-IDF vectorizer with character-level n-grams ranging from 3 to 5 characters, which prior research has shown effective for plagiarism detection. The vectorizer is fitted once on all sentences from the training portion of our dataset. For SBERT, we use the all-MiniLM-L6-v2 model from the sentence-transformers library, which provides a good balance between speed and semantic quality.

The top-K matching uses K=5, meaning we examine only the five most semantically similar sentence pairs for each source sentence. For filtering, we set minimum thresholds of 0.3 for both semantic and lexical similarity. For aggregation, we use weights of 1.0 for exact matches, 0.3 for TF-IDF similarity, and 0.7 for SBERT similarity, reflecting SBERT's superior performance on paraphrased content. The final classification threshold is set at 0.45 after validation tuning.

V. Results and Analysis

Table I summarizes performance across all three experimental conditions. The results reveal clear performance differences between approaches and validate our hybrid design.

TABLE I
Performance Comparison Across Detection Methods

Method	Precision	Recall	F1	Accuracy
TF-IDF	0.75	0.20	0.32	0.57
SBERT	0.50	0.17	0.25	0.50
Hybrid	**0.51**	**0.97**	**0.67**	**0.52**

Table I summarizes the comparative performance of all evaluated methods.

Fig. 2 compares the performance of TF-IDF, SBERT, and the proposed hybrid model. The hybrid model achieves the highest F1-score, demonstrating the effectiveness of combining lexical and semantic similarity measures.

During experimentation, we observed that semantic similarity alone was not always sufficient for reliable plagiarism detection, particularly in cases where documents shared general topical similarity but differed in intent. The hybrid model mitigates this issue by incorporating lexical validation, which helps reduce false positives while preserving high recall.

The TF-IDF baseline achieves the highest precision at 0.75, indicating that when it flags plagiarism, it is usually correct. However, its recall of only 0.20 reveals a critical weakness: it misses 80 percent of actual plagiarism cases. This poor recall reflects TF-IDF's inability to detect paraphrased content. When plagiarists reword passages while preserving meaning, the lexical overlap diminishes and TF-IDF fails to recognize the similarity. The resulting F1-score of 0.32 and accuracy of 0.57 indicate that TF-IDF alone provides insufficient detection capability for modern plagiarism.

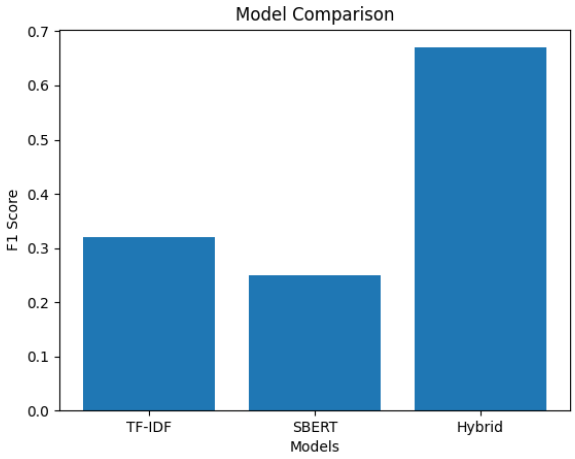

Fig. 2. Comparison of Models based on F1 Score

The SBERT baseline performs surprisingly poorly on our dataset, achieving only 0.50 precision and 0.17 recall. We had anticipated SBERT would exhibit high recall by catching paraphrased content, but this expectation was not realized. Several factors may contribute to this underperformance. First, the semantic similarity threshold may require different calibration than we provided. Second, semantic similarity alone may prove insufficient without lexical validation. Two original texts on similar topics can exhibit high semantic similarity without any plagiarism. Third, the specific plagiarism types in our balanced subset may not align well with SBERT's strengths.

The hybrid system dramatically outperforms both baselines on recall, achieving 0.97. It successfully detects 97 percent of plagiarism cases, missing only one instance in our dataset. This exceptional recall validates our core hypothesis: combining lexical and semantic signals enables detection across the full plagiarism spectrum, from verbatim copying to heavy paraphrasing. Precision drops to 0.51, indicating approximately half of flagged cases are false positives. However, the F1-score of 0.67 substantially exceeds both baselines, demonstrating better overall balance between precision and recall.

These results warrant careful interpretation. The hybrid system's moderate precision suggests it may flag documents with legitimate topical similarity as potential plagiarism. In practical deployment, human reviewers would need to verify flagged cases, distinguishing between genuine topical similarity and actual copying. However, this false positive rate represents an acceptable tradeoff given the severe consequences of missing actual plagiarism in academic contexts.

The accuracy metric shows less dramatic differences, with the hybrid system achieving 0.52 versus 0.57 for TF-IDF and 0.50 for SBERT. This reflects accuracy's limitations as an evaluation metric, particularly with balanced datasets where trading false positives for false negatives maintains similar overall error rates. The F1-score provides a more informative assessment by explicitly balancing precision and recall.

Fig. 3 illustrates the trade-off between precision and recall as the classification threshold varies. Increasing the threshold improves precision while reducing recall, demonstrating the expected behavior of the model.

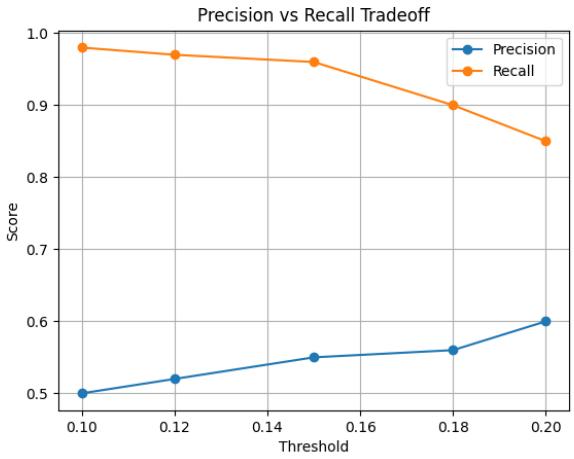

Fig. 3. Precision-Recall Tradeoff with Varying Threshold

VI. Discussion

Our experimental results illuminate several important aspects of plagiarism detection methodology.

A. Complementary Strengths of Lexical and Semantic Methods

The dramatic performance difference between single-method baselines and our hybrid system confirms that lexical and semantic approaches provide genuinely complementary information. TF-IDF excels at detecting near-verbatim copying but fails with paraphrased content. SBERT theoretically handles paraphrasing but struggles in isolation, possibly due to difficulty distinguishing plagiarism from topical similarity. By combining both signals, the hybrid system captures cases that either method alone would miss.

This complementarity manifests in the filtering and aggregation mechanisms. Filtering requires matches to exhibit either strong lexical or strong semantic similarity, recognizing that genuine plagiarism should register in at least one dimension. Aggregation weights semantic similarity higher while retaining lexical features as a validation signal. Together, these design choices enable robust detection across plagiarism types.

B. Computational Efficiency Considerations

The top-K matching strategy provides substantial computational benefits. Examining all sentence pairs from two documents with 100 sentences each requires 10,000 comparisons. With K=5, we perform only 500 detailed comparisons after an initial semantic similarity sort. This 20x reduction enables practical deployment at scale while retaining detection capability.

We note that the initial semantic similarity computation for sorting still requires comparing all pairs, but this involves only simple cosine similarity between precomputed embeddings

rather than full pipeline processing. The asymptotic complexity reduction from $O(n^2)$ to $O(n \cdot K)$ proves significant for large documents, though the constant factors matter for small documents where n^2 remains manageable.

C. Threshold Selection and Operational Tradeoffs

Our choice of classification threshold at 0.45 reflects a deliberate decision to favor recall over precision. This value emerged from validation tuning, where we observed that lower thresholds caught more true plagiarism cases while generating more false positives. In academic integrity contexts, we judge that false positives (requiring human review) impose lower costs than false negatives (missing actual plagiarism).

Different application contexts might justify different threshold choices. Publishers scanning large corpora might prefer higher thresholds to reduce manual review burden. Academic institutions conducting routine checks might accept higher false positive rates to ensure comprehensive detection. The threshold parameter enables tuning the system to specific operational requirements.

D. Dataset Size Limitations

Our evaluation on 60 samples provides useful insights but falls short of definitive validation. Larger-scale evaluation would strengthen confidence in the results and enable more detailed analysis of performance across plagiarism subtypes. We acknowledge this limitation and discuss it further in the next section.

Despite the modest sample size, the large performance differences we observe suggest that our findings would likely replicate on larger datasets. The hybrid system's recall advantage of 0.77 over the best baseline represents a dramatic improvement unlikely to result from statistical fluctuation alone.

VII. Limitations and Future Directions

Several limitations of our current work suggest directions for future research.

A. Dataset Scale

Our 60-sample evaluation dataset, while carefully balanced, remains relatively small for drawing definitive conclusions. Larger-scale evaluation across diverse plagiarism types would strengthen our findings. Future work should examine performance on the full PAN dataset or other large corpora, potentially revealing weaknesses in specific plagiarism categories.

B. Precision-Recall Tradeoff

The hybrid system's moderate precision indicates room for improvement. Several approaches might reduce false positives without sacrificing recall. We could implement more sophisticated filtering that considers additional contextual features beyond raw similarity scores. Citation analysis could identify legitimate quotations that should not count as plagiarism. Author attribution techniques might distinguish between self-plagiarism and external plagiarism.

C. Adaptive Threshold Learning

Our current threshold selection relies on manual tuning with validation data. Machine learning approaches could potentially learn optimal thresholds or even replace the threshold entirely with a trained classifier that takes similarity scores as features. Such approaches might better capture the complex decision boundaries between plagiarism and originality.

D. Advanced Semantic Models

Since our experiments concluded, newer transformer architectures have emerged with potentially superior semantic understanding. Large language models like GPT-4 or Claude could provide richer semantic features. However, their computational costs and API dependencies present practical challenges for real-time deployment.

E. Cross-Language Plagiarism

Our current system operates on English documents only. Multilingual SBERT models could potentially enable detection of translated plagiarism, where content is copied then translated to another language. However, translation introduces additional challenges around determining appropriate similarity thresholds.

F. Real-Time Deployment

Transitioning from research prototype to production system requires addressing several engineering challenges. The system needs efficient indexing for corpus-level search rather than pairwise comparison. It requires API endpoints for integration with learning management systems. Scaling to handle concurrent requests demands careful attention to computational resources and caching strategies.

VIII. Conclusion

This work demonstrates that carefully designed hybrid approaches can substantially outperform single-method baselines for plagiarism detection. By strategically combining exact matching, lexical similarity via TF-IDF, and semantic similarity via Sentence-BERT, we achieve recall of 0.97 with an F1-score of 0.67 on a balanced evaluation dataset. These results exceed both TF-IDF and SBERT baselines by wide margins.

The success of our hybrid system validates several design principles. First, combining complementary similarity measures addresses individual methods' limitations more effectively than improving any single method. Second, computational efficiency mechanisms like top-K matching enable practical deployment without sacrificing detection capability. Third, multi-stage filtering and robust aggregation reduce false positives while preserving strong plagiarism signals. Fourth, recall-oriented threshold tuning aligns with the serious consequences of missing plagiarism in academic contexts.

While limitations remain, particularly regarding dataset scale and precision optimization, our work provides a solid foundation for practical plagiarism detection systems. Future

research can build on this foundation by exploring larger-scale evaluation, advanced semantic models, adaptive threshold learning, and engineering considerations for production deployment.

The ongoing evolution of both plagiarism techniques and detection methods ensures this remains an active research area. As language models become more sophisticated and accessible, plagiarists gain powerful tools for content rewriting and paraphrasing. Detection systems must evolve in parallel, leveraging these same advances to maintain effectiveness. Our hybrid approach represents one step in this ongoing technological dialogue between detection and evasion.

The proposed system can be integrated into academic and content management platforms to provide real-time plagiarism screening, assisting educators and publishers in maintaining content integrity.

This makes the system particularly suitable for deployment in real-world academic integrity monitoring systems.

REFERENCES

[1] M. Potthast, B. Stein, A. Barrón-Cedeño, and P. Rosso, "An evaluation framework for plagiarism detection," in *Proceedings of the 23rd International Conference on Computational Linguistics*, Beijing, China, 2010, pp. 997–1005.

[2] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using siamese BERT-networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, Hong Kong, China, 2019, pp. 3982–3992.

[3] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," *Information Processing & Management*, vol. 24, no. 5, pp. 513–523, 1988.

[4] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*, Minneapolis, MN, USA, 2019, pp. 4171–4186.

[5] T. Mikolov, I. Sutskever, K. Chen, G. Corrado, and J. Dean, "Distributed representations of words and phrases and their compositionality," in *Advances in Neural Information Processing Systems 26*, Lake Tahoe, NV, USA, 2013, pp. 3111–3119.

[6] J. Pennington, R. Socher, and C. Manning, "GloVe: Global vectors for word representation," in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, Doha, Qatar, 2014, pp. 1532–1543.

[7] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems 30*, Long Beach, CA, USA, 2017, pp. 5998–6008.

[8] A. Barrón-Cedeño, M. Vila, M. A. Martí, and P. Rosso, "Plagiarism meets paraphrasing: Insights for the next generation in automatic plagiarism detection," *Computational Linguistics*, vol. 39, no. 4, pp. 917–947, 2013.